AN ETALE MORPHISM OF ALGEBRAIC SPACES IS LOCALLY QUASI-FINITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement is one sentence; the proof is the standard presentation reduction naming the definition, the locality statement and the scheme case.

Editorial note. The four references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The definition of etaleness for morphisms of algebraic spaces. Properties of Spaces, Definition spaces-properties-definition-etale, used at proof lines 7983-7984. Fixes the property being used.
- The source-and-target presentation for etale morphisms. Properties of Spaces, Lemma spaces-properties-lemma-etale-local and Lemma lemma-local-source-target, used at proof lines 7985-7989. Produces the scheme morphism h.
- The scheme case: an etale morphism is locally quasi-finite. Morphisms, Lemma morphisms-lemma-etale-locally-quasi-finite, used at proof lines 7989-7991. Gives local quasi-finiteness of h.
- Local quasi-finiteness read off from the presentation. used at proof lines 7991-7992. Concludes for X -> Y.

Source reference. Original lemma 03MJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4751–4751. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03WS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5123–5123. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original definition 03FR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4502–4502. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03FS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4503–4503. Navigation pointer into the Stacks

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Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *An étale morphism of algebraic spaces is locally quasi-finite.*

Proof. Let $X \to Y$ be an étale morphism of algebraic spaces, see Properties of Spaces, Definition 03FR. By Properties of Spaces, Lemma 03FS we see this means there exists a diagram as in Lemma 03MJ with h étale and surjective vertical arrow a. By Morphisms, Lemma 03WS h is locally quasi-finite. Hence $X \to Y$ is locally quasi-finite by definition. □